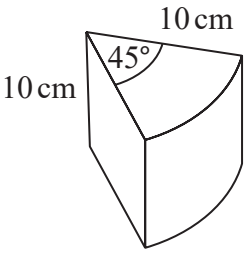
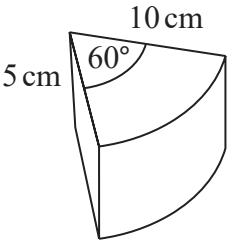
19 Here are two solid prisms, prism **A** and prism **B**.



prism **A**



prism **B**

The cross section of prism **A** is a sector, with angle 45°, of a circle of radius 10 cm.
The prism has a depth of 10 cm and a mass of 40π grams.

The cross section of prism **B** is a sector, with angle 60°, of a circle of radius 10 cm.
The prism has a depth of 5 cm and a mass of 50π grams.

Express the difference in the densities of the two prisms as a percentage of the density of prism **A**.

A volume = $(45 \div 360) \times \pi \times 10^2 \times 10 = 125\pi$

1 mark
Sector volumes.

B volume = $(60 \div 360) \times \pi \times 10^2 \times 5 = \dfrac{250\pi}{3}$

A density = $40\pi \div 125\pi = 0.32\ g/cm^3$

1 mark
Mass ÷ volume.

B density = $50\pi \div (250\pi \div 3) = 0.6\ g/cm^3$

1 mark
Both densities.

Difference = $0.6 - 0.32 = 0.28\ g/cm^3$

1 mark
Compare with A.

Percentage = $0.28 \div 0.32 \times 100 = 87.5\%$

1 mark
87.5%.

..... %

(Total for Question 19 is 5 marks)